



Health-awareness energy management strategy for battery electric vehicles based on self-attention deep reinforcement learning

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HIGHLIGHTS

- The health awareness of multiple electric components is embedded in EMS.
- The GRU is employed to improve the performance of the EMS after convergence.
- The Self-attention mechanism is adopted to improve the adaptability of EMS.
- The proposed EMS outperforms the other EMSs in both training and testing cycles.

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ABSTRACT

The economical and safe energy management strategy (EMS) is essential for a battery electric vehicle (BEV). With an improved deep deterministic policy gradient (DDPG)-based EMS proposed for a dual-motor BEV, energy consumption and electric component aging are minimized. Firstly, this study incorporates health awareness of electric components into EMS. Then, this study employs a gated recurrent unit (GRU) as well as a self-attention (SA) mechanism for improving the original DDPG's performance. The GRU enhances the performance of DDPG agents, post-convergence, by furnishing them with additional state information. The SA mechanism empowers agents to distinguish the importance of state information, thus improving its adaptability in new environments. The proposed EMS decreases battery energy consumption to 7.95 kW-hour and enhances battery state of health (SOH) to 96.83 % of the dynamic programming (DP)-based EMS. The two motors of the proposed EMS possess the highest end-value SOH, i.e., 99.99929 % and 99.99978 %, respectively. The proposed EMS demonstrates superior adaptability to other EMSs, regardless of the initial state of charge (SOC). It shows the best adaptability at an initial SOC of 0.6, and achieves 95.87 % and 94.67 % of the energy consumption of DP-based EMS in the mixed and actual test cycles, respectively.

1. Introduction

Under the goal of sustainable transportation development, the automobile industries of major global automobile countries have embarked on the transformation path of green development. Battery electric vehicles (BEVs) are becoming the main direction of the transformation and development of the global automobile industry. In comparison to traditional fuel powered vehicles, the advantages of BEVs include zero emissions, low noise, and fast acceleration. However, short mileage and low motor efficiency have hindered the popularization of BEVs worldwide [1].

The powertrain of most BEVs available on the market is mainly based on the "single motor + single gear reducer", which is simple to manufacture and has low manufacturing costs. To meet the dynamic requirements for motor torque during low-speed starting and high-speed overtaking, the selected drive motors often have a large spare power, which results in the motor's operating point deviating from the high-efficiency region and seriously wasting the motor's spare torque during most of the working time [2]. In addition, coupling the drive motor to the wheels results in a "single motor + single gear reducer" powertrain that cannot adjust the motor's operating point in real-time according to different road conditions, thus further exacerbating the

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driver's mileage anxiety. Therefore, to meet the dual demand for dynamic performance and economic performance of BEVs, and to solve the problem that BEVs cannot balance the contradiction between dynamic and economy under a single power source, dual-motor drive BEVs have gradually been favored by the market and become a research hotspot at home and abroad [3].

Nomenclature Symbols

Abbreviations:			
EMS	energy management strategy	a_y	acceleration
BEV	battery electric vehicle	g	gravity acceleration
HEV	hybrid electric vehicle	φ	slope angle
DDPG	deep deterministic policy gradient	f	rolling resistance factor
GRU	gated recurrent unit	R	wheel radius
SA	self-attention	v	velocity
SOH	state of health	A	frontal area
SOC	state of charge	C_d	air resistance factor
DP	dynamic programming	I	load current
RL	reinforcement learning	Q_{bat}	nominal capacity
DRL	deep reinforcement learning	V_{bi}	polarization voltage
DQL	deep Q-learning	V_t	terminal voltage
DDQN	double deep Q-network	V_{oc}	open circuit voltage
MLP	multiplayer perceptron	C_{bi}	capacitances of two RC branches
LSTM	long short-term memory	R_{bi}	resistance of two RC branches
BMS	battery management system	R_s	ohmic resistance
BTMS	battery temperature management system	T_s	temperature of battery surface
DC-DC	direct current to direct current	T_c	temperature of battery core
EOL	end of life	T_a	average internal temperature
CLR	cumulative loss ratio	T_f	ambient temperature
AN	actor-network	R_c	convection at the battery surface
CN	critic-network	R_{ui}	heat conduction inside the cell
NEDC	new European driving cycle	C_c	battery thermal capacitances in core
WLTC	world light vehicle test cycle	C_s	battery thermal capacitances in surface
UDDS	urban dynamometer driving schedule	H_t	heat generation rate
HWFET	highway fuel economy test	E_n	entropy variation
CLTC-P	China light-duty vehicle test cycle-passenger	ΔQ_{bat}	loss of battery capacity
FTP75	federal test procedure 75	c	C-rate
Symbols:		B	pre-exponential coefficient
δ	conversion coefficient for rotational mass	R_g	ideal gas constant
m	vehicle mass	Ah	cumulative amp-hour throughput
E_a	activation energy	z	power-law coefficient
W_{losses}	total energy loss of motor	N	overall number of cycles
P_{rated}	rated power of motor	t_{life}	life cycle of the motor
W_{loss_i}	energy loss of the i^{th} run	CLR	cumulative loss ratio
$t_{initial_i}$	start time of the i^{th} run	η_i	motor efficiency of the i^{th} run
t_{final_i}	end time of the i^{th} run	y_i	target Q-value
P_{in_i}	input power of the i^{th} run	Q	network parameter for CN
P_{out_i}	output power of the i^{th} run	μ	network parameter for AN
ξ	factor that controls the soft update rate	M	mini-batch
W_r	weight of reset gate	$\nabla_{\theta'} J$	loss function for CN
W_z	weight of update gate	$\nabla_{\theta} J$	loss function for AN
b_r	bias for update gate	Q'	network parameters for target CN
b_z	bias for reset gate	μ'	network parameters for target AN
x_t	current input	W_h	candidate state weight
h_{t-1}	previous hidden state	b_h	bias of candidate state
d_Q	scaling factor	h_t	current hidden state
h_t^c	memory pool contains historical information	$\tilde{h}_t$	candidate current hidden state

(continued on next column)

(continued)

L	sequence length of historical state information	Q	query
K_i	positive factors	K	key
σ_{tor}	torque distribution factor	V	value
a^0	zero-valued virtual action vectors	H_s	output of SA
s^0	zero-valued virtual observation vectors		

On the other hand, dual-motor drive BEVs must be equipped with precise energy management strategy (EMS) for coordinating dual motors control to improve the energy utilization efficiency of the entire vehicle, which can make up for the single-motor drive BEVs cannot consider the dynamic and economic performance at the same time. According to the current proposals, EMSs fall into three categories [4]: rule-based, optimization-based, and learning-based. EMSs based rule are designed easily and applied in real-time. However, developing rules depends on the engineering experience of experienced experts. In addition, predefined rules cannot effectively ensure the robustness and adaptability of rule-based EMSs under complex or unknown driving conditions [5]. The objective function is established and constraints are selected to cast around for the optimal EMSs based on the optimization problem. Among them, the dynamic programming (DP) approach is typical of global optimal-based EMSs [6]. However, applying global optimization-based EMSs in practice is challenging due to the need for prior driving condition information and extensive computation. Instantaneous optimization-based EMSs, on the other hand, optimize energy consumption without future driving condition information, relying solely on currently available driving condition data. However, a comparison between instantaneous optimization EMSs and global optimization EMSs shows that instantaneous optimization EMSs are still capable of improving their performance [7].

With the fast progress of machine learning, EMSs are also advancing toward intelligence. Reinforcement learning (RL)-based EMSs are the current research hotspot. These EMSs utilize RL to train the intelligent agent, which can then be applied to real-time online learning after sufficient training [8]. This approach offers a viable solution for challenging issues beyond the capabilities of EMSs based on rules and optimization theory, e.g. heavy computational loading and terrible adaptability [9]. Reference [10] utilized a time-varying exploration algorithm instead of the ϵ -greedy algorithm in traditional Q-learning to improve the real-time performance of Q-learning-based EMS. However, to store the Q-values for every state-action combination, Q-Learning uses tables, which causes the "curse of dimensionality" in high-dimensional state and action spaces to be an issue for Q-Learning-based EMSs [11]. Deep Q-learning (DQL), which uses RL and neural networks to match the value function and policy function and make up for the shortfalls of the Q-learning algorithm, is a solution to this restriction [12,13]. Consequently, DQL is introduced into EMSs. Reference [14] compared Q-Learning-based and DQL-based EMSs in a hybrid battery system for BEVs, revealing that the latter outperformed the former in computation time and energy consumption. However, the optimistic estimation of Q-value by DQL, which causes the production of suboptimal EMSs. Double deep Q-network (DDQN) overcomes this issue by utilizing two neural networks to separate action selection and Q-value evaluation. Therefore, reference [15] designed a DDQN-based EMS for electric-hydraulic hybrid electric vehicle (HEV), which achieved optimal switching of working modes and significantly improved vehicle economy. Unfortunately, DQL and DDQN only support discrete action spaces. To cope with continuous action spaces, researchers have begun to apply the deep deterministic policy gradient (DDPG) method to EMSs using an actor-critic framework. A DDPG-based EMS with a unique incentive system for a dual-motor, dual-axle-drive BEV was proposed by Cui et al. [3], achieving higher quality and faster learning compared to the DDQN-based EMS. To further enhance vehicle mileage, Ruan et al.

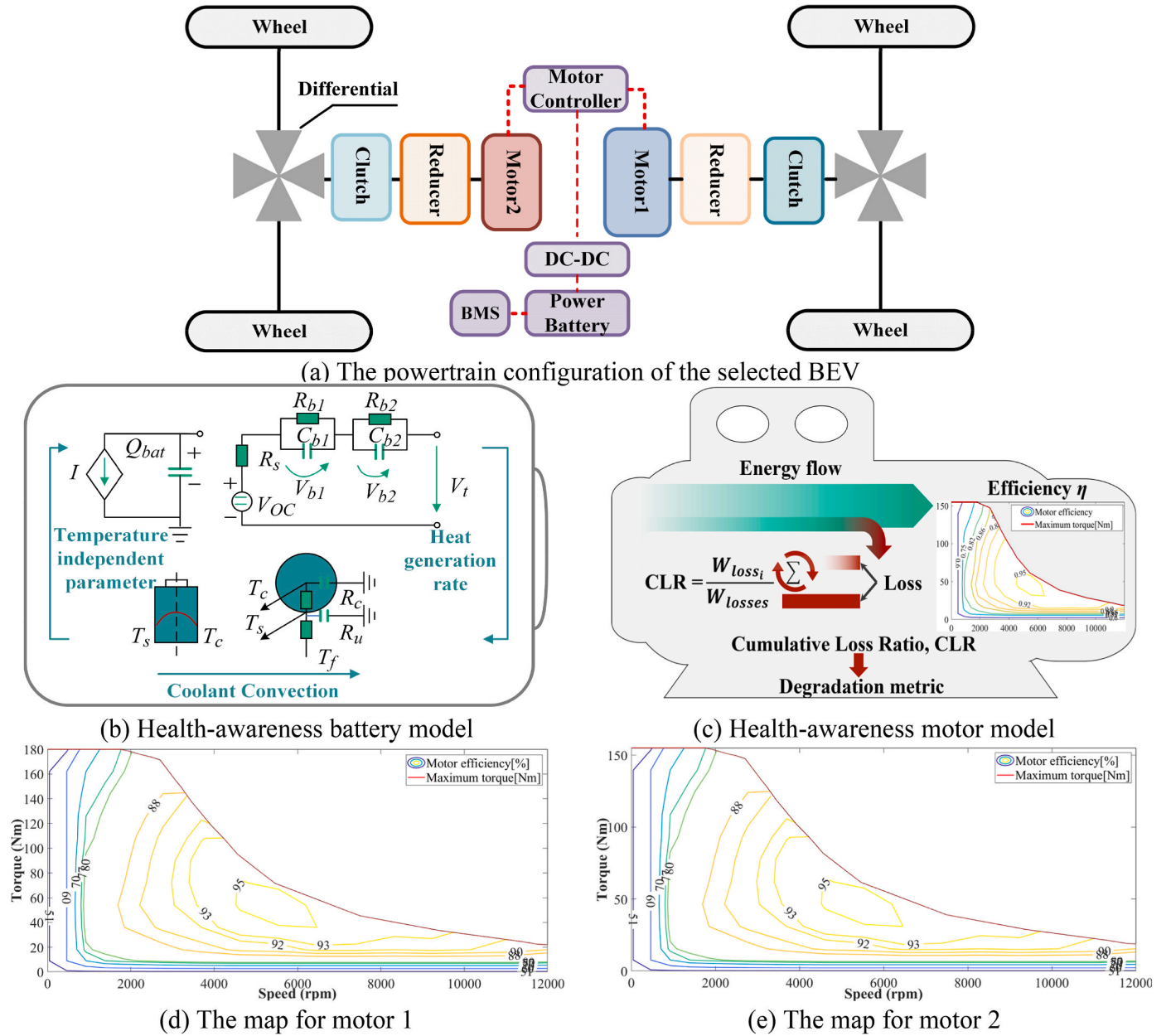


Fig. 1. The powertrain configuration of the selected BEV.

[16] embedded the braking energy recovery strategy and driving mode determination algorithm into the DDPG-based EMS, forming a hierarchical EMS. Lithium-ion batteries are preferred for their high energy density and longevity [17], but they can experience temperature rises, capacity loss, and thermal runaway at high discharge rates [18]. To address these issues, researchers improved battery temperature management system (BTMS) through innovative structural designs [19,20] and materials [21–23]. On this basis, considering that battery played a crucial role in BEVs, Lu et al. [24] constructed a battery health-aware EMS based on DDPG, which increased the average efficiency of the battery by 2.02 % and reduced the average performance degradation of the battery by 14.4 %.

Despite making significant progress in RL-based EMSs, three primary drawbacks require further improvement. Firstly, Multiple motor driving is becoming more common in BEVs, which enhances the overall economy and performance of the vehicle [25]. As the core components of BEVs, the motors' performance and lifespan directly affect the vehicle's performance and mileage. The addition of extra motors brings

challenges to the precise control of motors in BEVs. However, few RL-based EMSs currently take the state of health (SOH) of the motors into account. Secondly, most of the adopted RL algorithms' networks consist of multiplayer perceptron (MLP) containing only activation functions and linear activation functions [26], which limits the performance of the model after convergence. Although there are already some studies to improve agent performance by embedding long short-term memory (LSTM) networks into the actor-network, the complex gate structure of LSTM limits improvement in agent learning speed. Thirdly, unlike human drivers who selectively focus on environmental state information to make decisions, the current RL-based EMSs pay equal attention to all input state information. The lack of discrimination results in the RL-based EMSs having poor adaptability in new environments.

This study proposes the improved EMS for the above research gaps, which controls electrical energy consumption while minimizing battery and motors' aging. The three contributions that this work seeks to make to the present deep reinforcement learning (DRL)-based EMSs for BEVs

Table 1

The main parameters of BEV.

Description	Parameter	Value	Unit
Vehicle Weight	m	1635	kg
Front Area	A	2.16	m^2
Rolling Resistance Factor	f	0.016	–
Air Resistance Factor	C_d	0.25	–
Wheel Radius	R	0.325	m
Battery Energy	E_{bat}	128	kWh
Front Reducer Gear Ratio	k_1	11	–
Rear Reducer Gear Ratio	k_2	6	–
Motor1 Max Torque	$T_{1\ max}$	180	Nm
Motor1 Max Speed	$N_{1\ max}$	12000	rpm
Motor1 Rated power	P_{1_rated}	28	kW
Motor1 Rated Efficiency	η_{1_rated}	0.98	–
Motor2 Max Torque	$T_{2\ max}$	155	Nm
Motor2 Max Speed	$N_{2\ max}$	12000	rpm
Motor2 Rated power	P_{2_rated}	24	kW
Motor2 Rated Efficiency	η_{2_rated}	0.98	–

are as follows:

- 1) By incorporating health awareness of critical electrical components into the EMS of the BEV, the need for precise control of critical electrical components is met, slowing down the costly degradation process.
- 2) The DDPG agent incorporates the gate recurrent unit (GRU) to form recurrent actor-critic networks. This enables the agent to observe more comprehensive state information, leading to improved performance after convergence. Additionally, the GRU's simple internal structure accelerates the agent's learning process.
- 3) By introducing the self-attention (SA) mechanism, the constructed recurrent DDPG agent can be more flexible in choosing the state information to focus on, resulting in improved performance and expressiveness of the model, enhancing its adaptability in different environments.

This work is structured as follows. In Section 2, a health-awareness BEV powertrain model is presented. The EMS is developed in Section 3 using the improved DDPG. The findings of the simulation are examined and analyzed in Section 4. In Section 5, the study's shortcomings are outlined and future directions are suggested.

2. Model description

2.1. Vehicle modeling

Fig. 1(a) depicts the powertrain of the BEV chosen in this study. The power battery is the only energy source for the BEV. The battery management system (BMS) manages the battery's charging and discharging while monitoring its status. The DC-DC converter lowers the battery's voltage to power the BEV's electrical appliances. The motor controller governs the power outputs of motor 1 and motor 2, allowing the BEV to switch between two-wheel and four-wheel drive by controlling the motors and clutch. The vehicle parameters derived from Ref. [2] are presented in Table 1.

The required driving torque T_d of the BEV is formulated by longitudinal dynamics [4] as follows:

$$T_d = \left(\delta m a_v + mg \sin \varphi + mgf \cos \varphi + \frac{C_d A v^2}{21.15} \right) R \quad (1)$$

where δ denotes the conversion coefficient for rotational mass; m denotes the mass of the vehicle; a_v is the driving acceleration; The slope angle is represented by φ , the rolling resistance factor by f , the gravity acceleration by g , and the factor of air resistance by C_d ; The vehicle's wheel radius is R , its velocity is v , and its frontal area is A .

2.2. Health-awareness modeling of battery

The power battery serves as the foundation of a BEV's energy power system, with its SOH playing a critical role in the vehicle's overall performance, including dynamics, economy, safety, and service life. As a result, there is a need to integrate health awareness into battery modeling. This study employs a coupled electrical-thermal-aging model to simulate the power battery system, where three sub-models exist, as shown in Fig. 1(b). The technical specification of the battery cell is as follows: the nominal voltage is 3.3 V, the nominal capacity is 7.6 Wh, the anode and cathode materials are graphite and lithium iron phosphate respectively, and the electrolyte material is lithium hexafluorophosphate.

The electrical and thermal models are represented by (2) and (3) [27], respectively.

$$\left\{ \begin{array}{l} \frac{dSOC(t)}{dt} = \frac{I(t)}{3600Q_{bat}} \\ \frac{dV_{b1}}{dt} = -\frac{V_{b1}(t)}{R_{b1}(t)C_{b1}(t)} + \frac{I(t)}{C_{b1}(t)} \\ \frac{dV_{b2}}{dt} = -\frac{V_{b2}(t)}{R_{b2}(t)C_{b2}(t)} + \frac{I(t)}{C_{b2}(t)} \\ V_t(t) = V_{oc}(SOC, t) + V_{b1}(t) + V_{b2}(t) + R_s I(t) \end{array} \right. \quad (2)$$

$$\left\{ \begin{array}{l} \frac{dT_s(t)}{dt} = \frac{T_s(t)}{R_u C_s} - \frac{2T_s(t)}{R_c C_s} + \frac{2T_a(t)}{R_c C_s} + \frac{T_f}{R_u C_s} \\ \frac{dT_a(t)}{dt} = \left(\frac{C_s - C_c}{R_c C_c C_s} - \frac{1}{2R_u C_s} \right) T_s(t) + \frac{C_c - C_s}{R_c C_c C_s} T_a(t) + \frac{H_t(t)}{2C_c} + \frac{T_f}{2R_u C_s} \\ T_c(t) = 2T_a(t) - T_s(t) \end{array} \right. \quad (3)$$

where SOC is the battery state of charge; I stands for the load current respectively; Q_{bat} is the nominal capacity; V_{b1} and V_{b2} denote the polarization voltage; V_t stands for terminal voltage; V_{oc} is the battery open circuit voltage; C_{b1} & C_{b2} are the capacitances of two RC branches; R_{b1} & R_{b2} are the resistance of two RC branches; R_s is the ohmic resistance. T_s is the temperature of battery surface; T_c is temperature of battery core; T_a is the average internal temperature; T_f is ambient temperature; R_c & R_u represent the convection at the battery surface and the heat conduction inside the cell, respectively; C_c & C_s are equivalent battery thermal capacitances in core and surface, respectively. Specific values of the above parameters can be found in Ref. [28]. H_t symbolizes the heat generation rate, which is computed using (4) [29], and E_n represents the entropy variation that happens during electrochemical processes.

$$H(t) = I(t)(V_{b1}(t) + V_{b2}(t) + R_s I(t)) + I(t)(T_a(t) + 273.15)E_n(SOC, t) \quad (4)$$

The energy-throughput aging model postulates that a battery can sustain a given amount of energy throughput under continuous operating circumstances prior to reaching its end of life (EOL), which equates to several cycles of charge and discharge.

The loss of battery capacity $\Delta Q_{bat}(\%)$ is calculated by Arrhenius equation, as shown in the (5) [8]:

$$\Delta Q_{bat} = B(c) \bullet e^{\frac{-E_a(c)}{R_g T_a}} \bullet Ah(c)^z \quad (5)$$

where C-rate is denoted by c ; $B(c)$ is the pre-exponential coefficient; R_g stands for ideal gas constant and its value is $8.31 \text{ J/mol} \bullet K$; The cumulative amp-hour throughput is denoted by Ah ; Power-law coefficient z is 0.55; Activation energy $E_a(\text{J/mol})$ is calculated by (6) [29].

$$E_a(c) = 31700 - 370.3 \bullet c \quad (6)$$

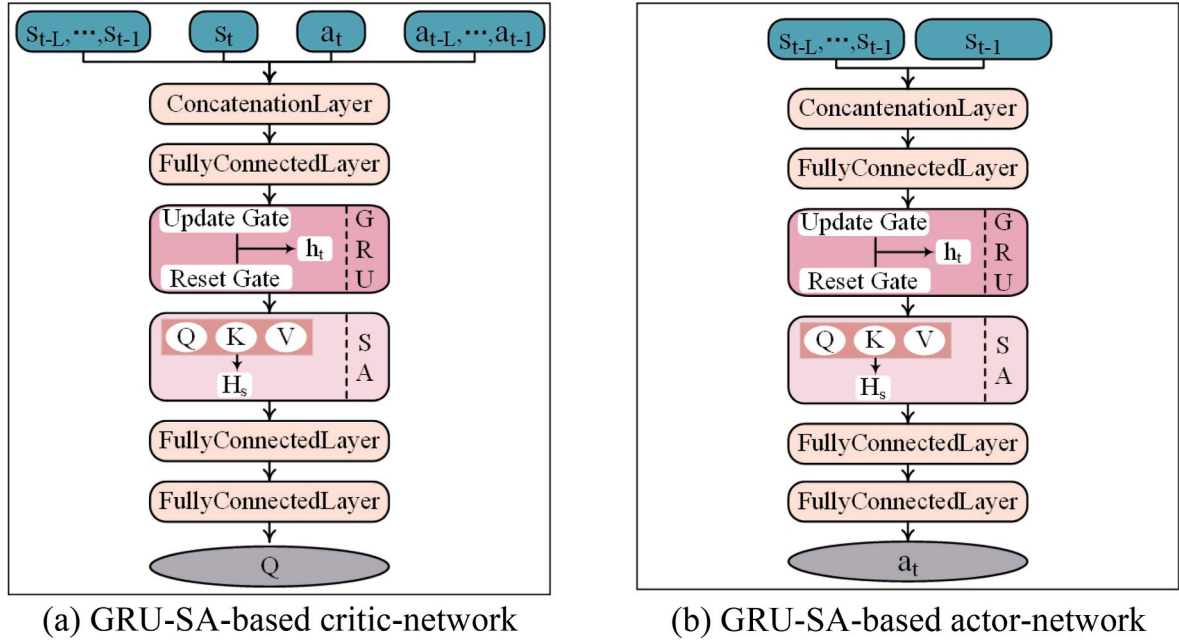


Fig. 2. Improved network structure of DDPG.

Table 2

Hyperparameter settings.

Parameter Description	Value
Number of hidden layers of Actor and Critic (target) network	3,3
Neurons distribution of hidden layers for Actor and Critic (target) network	64, 64, 64
Neurons distribution of input layers for Actor and Critic (target) network	6,7
Neurons distribution of output layers for Actor and Critic (target) network	1,1
Neurons distribution of GRU layer	50
Sequence length L	2
Optimizer	Adam
Discount factor	0.99
Learning rate of critic	1e-5
Learning rate of actor	1e-4
Exploration noise	0.35
Size of experience replay buffer	1e6
Minibatch size	50
Number of training episodes	500

The EOL is reached when the Q_{bat} of the power battery drops by 20 %. Based on this principle, (5) can be rewritten as follows [30].

$$Ah(c, T_a) = \left(\frac{20}{B(c) \cdot \exp\left(\frac{-E_a(c)}{R_g(T_a + 273.15)}\right)} \right)^{\frac{1}{\alpha}} \quad (7)$$

Then, (8) determines the overall number of cycles N until reaching EOL [31]:

$$N(c, T_a) = \frac{3600Ah(c, T_a)}{Q_{bat}} \quad (8)$$

Thus, the decline in the SOH under multiple stresses is calculated by (9) [32]:

$$\Delta SOH = -\frac{I(t) \Delta t}{2N(c, T_a) Q_{bat}} \quad (9)$$

2.3. Health-awareness modeling of motors

Motor 1 and Motor 2 in this study are modeled as quasi-static models,

and their relevant data is obtained from Ref. [2]. Both motors are permanent magnet synchronous motors. The power loss maps for the two motors can be found in Fig. 1(d) and (e).

One of the essential parts of BEVs is the motor, which together with the battery gives the vehicle efficient driving power. Consequently, it is also essential to incorporate health awareness into the motor modeling process. Consequently, the degradation cost function, i.e., cumulative loss ratio (CLR), is employed to define the degree of degradation of the motor.

The total energy loss over the life cycle of a motor with rated efficiency η_{rated} , if operated continuously, is defined by (10) [33]. However, the motor runs intermittently. Assuming that the motor is run intermittently for T cycles, the motor energy loss for the i^{th} ($0 \leq i \leq T$) run is calculated by (11) [33].

$$W_{losses} = \int_0^{t_{life}} \left(\frac{1}{\eta_{rated}} - 1 \right) P_{rated} dt \quad (10)$$

$$W_{loss_i} = \int_{t_{initial_i}}^{t_{final_i}} P_{in_i} - P_{out_i} dt = \int_{t_{initial_i}}^{t_{final_i}} P_{out_i} (1 - \eta_i) dt \quad (11)$$

where W_{losses} is the energy loss of the motor during continuous operation; t_{life} is the life cycle of the motor and its value is 30,000 h; P_{rated} is rated power; W_{loss_i} is the energy loss of the i^{th} run; $t_{initial_i}$ and t_{final_i} are the start time and end time of the i^{th} run, respectively; P_{in_i} and P_{out_i} are the input and output power of the i^{th} run; η_i is the motor efficiency of the i^{th} run.

The CLR ought to be considered in deriving degradation metric. The CLR is defined by (12) [29]:

$$CLR = \frac{\sum_{i=1}^T W_{loss_i}}{W_{losses}} \quad (12)$$

Specifically, the portion of energy loss that the motor is designed to tolerate is represented by CLR, which means that if this value is surpassed, there will be a loss of performance. Thus, the motor health status SOH_{mot} is calculated by (13) [34].

$$SOH_{mot} = 1 - CLR \quad (13)$$

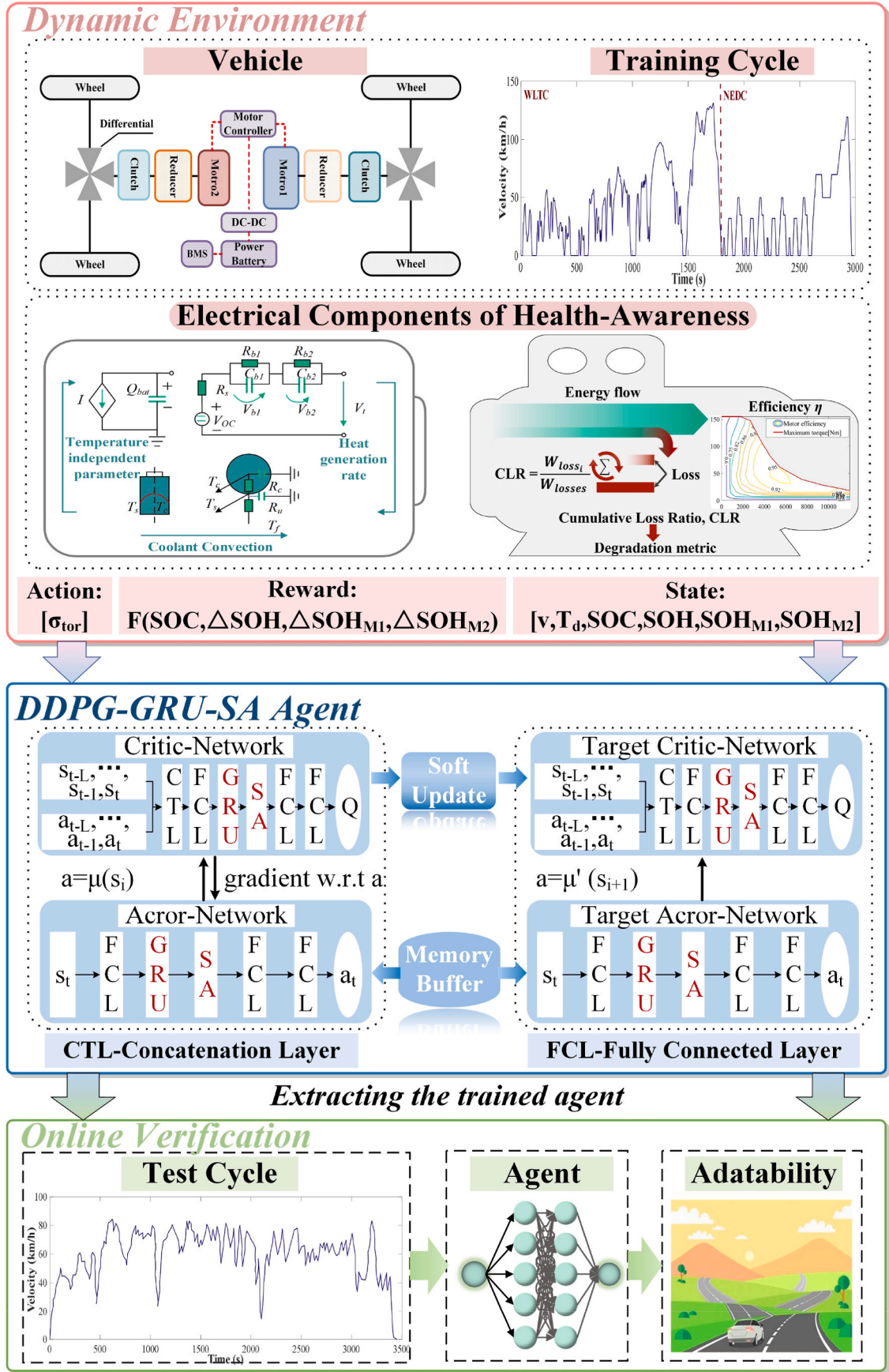


Fig. 3. The implementation framework of proposed EMS.

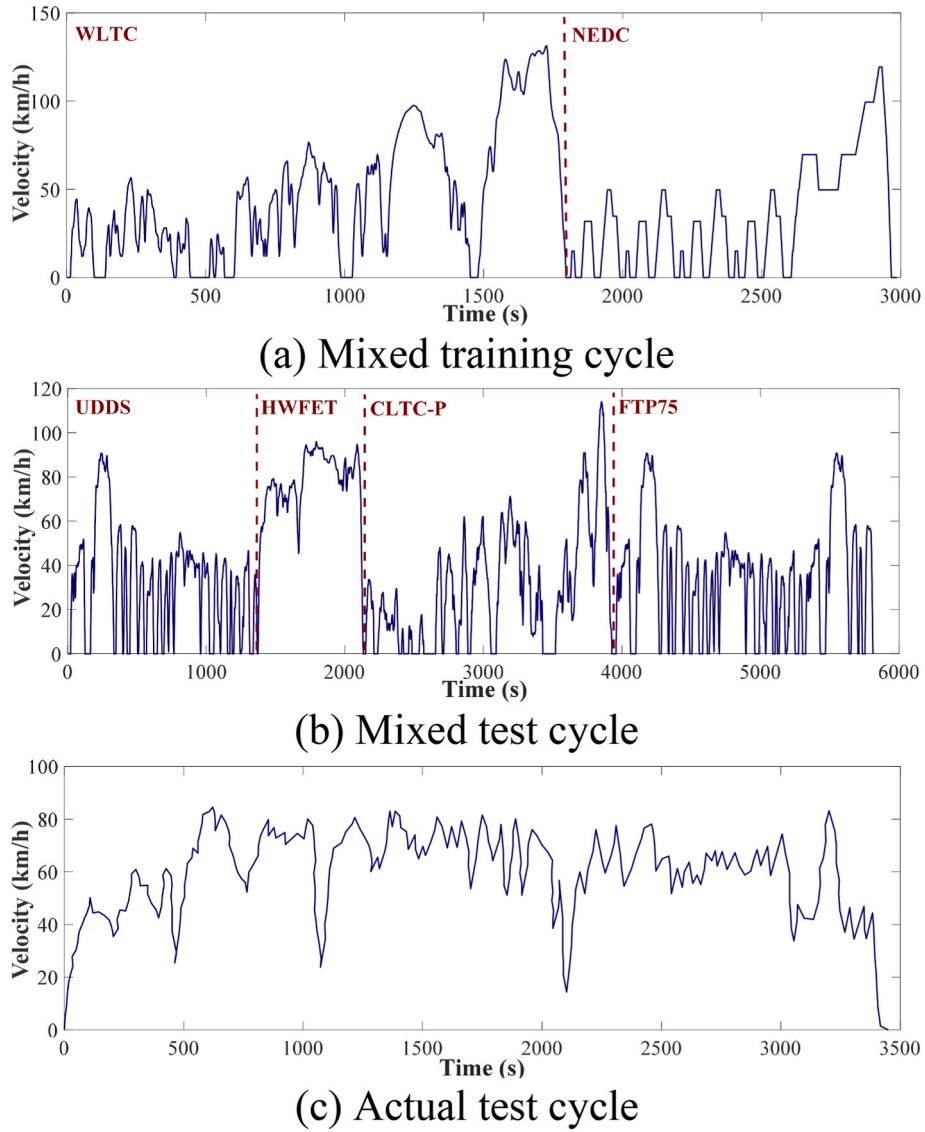


Fig. 4. The mixed driving cycles.

As shown in (11) to (13), the CLR can be decreased and thus the SOH_{mot} can be increased by improving the efficiency of the motor operation. The motor model with health awareness is drawn in Fig. 1(c).

3. Methods of designing EMS

3.1. Theory of the adopted methods

3.1.1. Introduction to DDPG

DDPG method consists of four neural networks: the actor-network (AN), the critic-network (CN), the target AN, and the target CN. The CN and AN are updated by (14) and (15) [35], respectively:

$$\nabla_{\theta^Q} J = \frac{1}{M} \sum_{i=1}^M (y_i - Q(s_i, a_i | \theta^Q))^2 \quad (14)$$

$$\nabla_{\theta^\mu} J = \frac{1}{M} \sum_i [\nabla_{\theta^Q} Q(s_i, a_i | \theta^Q) \nabla_{\theta^\mu} \mu(s_i | \theta^\mu)] \quad (15)$$

where y_i stands for the target value of $Q(s, a)$; Q and μ represent the network parameters for CN and AN, respectively. M is mini-batch; $\nabla_{\theta^Q} J$ and $\nabla_{\theta^\mu} J$ are loss function for CN and AN, respectively.

The target AN and CN can be updated by (16) and (17) [3]:

$$\theta^Q \leftarrow \xi \theta^Q + (1 - \xi) \theta^{Q'} \quad (16)$$

$$\theta^\mu \leftarrow \xi \theta^\mu + (1 - \xi) \theta^{\mu'} \quad (17)$$

where ξ is factor that controls the soft update rate of the target networks, and $\xi \ll 1$; Q' and μ' are the network parameters for target CN and AN, respectively.

3.1.2. Introduction to GRU

GRU method simplifies the gate structure of LSTM into two gates (reset gate r_t and update gate z_t), making it easier to train and converge faster. The computational process of GRU is controlled by (18) [36].

$$\begin{cases} r_t = \text{sigmoid}(W_r \bullet [h_{t-1}, x_t] + b_r) \\ z_t = \text{sigmoid}(W_z \bullet [h_{t-1}, x_t] + b_z) \\ \bar{h}_t = \tanh(W_h \bullet [r_t \bullet h_{t-1}, x_t] + b_h) \\ h_t = (1 - z_t) \bullet \bar{h}_t + z_t \bullet h_{t-1} \end{cases} \quad (18)$$

where W_r and W_z are weights of reset & update gates, respectively; W_h is candidate state weight; b_r and b_z are the biases for update & reset gates,

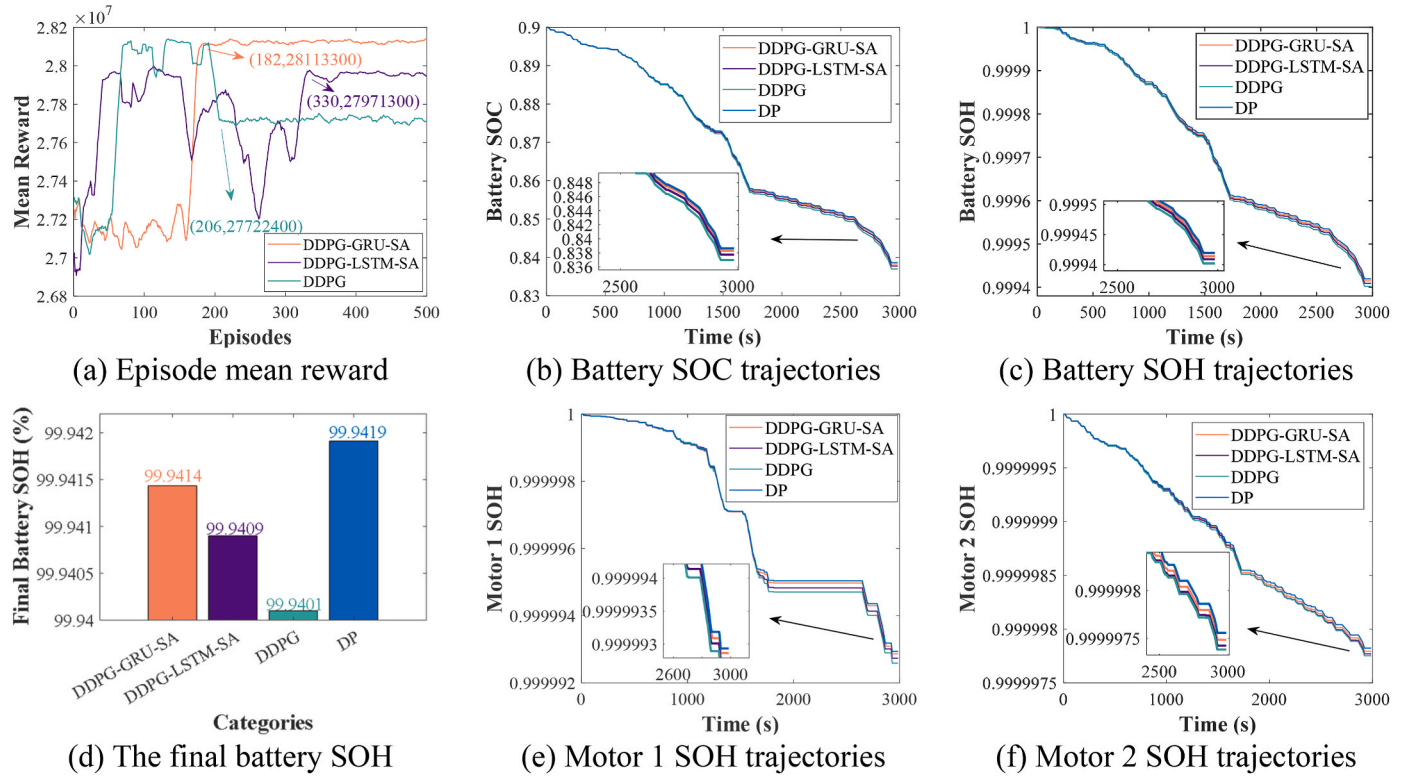


Fig. 5. Performance comparison of different EMSs.

Table 3
Comparative analysis between different EMSs.

EMSs	Running time (s)	Electricity consumption (kWh)	Performance	Battery decline	Performance	Motors' decline	Performance
DP	42580	7.71	100 %	5.68e-4	100 %	0.92e-6	100 %
DDPG-GRU-SA	31520	7.95	96.88 %	5.86e-4	96.83 %	0.94e-6	97.83 %
DDPG-LSTM-SA	35230	8.03	95.85 %	5.91e-4	95.95 %	0.96e-6	95.65 %
DDPG	40123	8.13	94.55 %	5.99e-4	94.54 %	0.99e-6	92.40 %

respectively; b_h is bias of candidate state; x_t is current input; h_t is current hidden state; h_{t-1} is previous hidden state; $\bar{h}_t$ is candidate current hidden state.

3.1.3. Introduction to SA mechanism

The SA mechanism is superior at capturing the internal importance of data or characteristics, can link information at different points in the input sequence, and lessens the need on external information. Three components make up the attention mechanism: query, key, and value. The query and key for the SA mechanism show the identical input signal sequence. Equation (19) expresses the SA mechanism's output H_s [37]:

$$H_s = \text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_q}}\right)V \quad (19)$$

where $\sqrt{d_q}$ denotes scaling factor; Q denotes the query; K denotes the key; V denotes the value.

3.2. DDPG in combination with GRU and SA mechanism

The simple structure of the MLP results in a weaker model representation. The performance of the original DDPG employing MLP as the actor and critic networks is limited, i.e., the training time is too long and prone to fall into local optima.

As seen in Fig. 2, this study integrates the GRU and SA mechanism into the AN and CN of DDPG in order to overcome the above problems.

As mentioned in the [38], energy management is more deserving to be modeled as the partially observable Markov decision processes. Unlike the original DDPG which can only observe the current state information, the DDPG agent embedded with GRU acquires more state information, i.e., current state information and historical state information, which improves the convergence performance of the DDPG. Moreover, unlike previous studies that embedded LSTM into DDPG [39], the simpler structure of GRU accelerates DDPG agent learning.

After embedding the GRU, the mini-batch $M\{(s_t, a_t, r_t, s_{t+1})\}$ sampled from the memory pool additionally contains historical state information h_t^L which is defined as (20) [28], so mini-batch is rewritten as $M\{(h_t^L, s_t, a_t, r_t, s_{t+1})\}$.

$$h_t^L = \begin{cases} \{s_{t-L}, a_{t-L}, \dots, s_{t-2}, a_{t-2}, s_{t-1}, a_{t-1}\}, & t \text{ and } L \geq 1 \\ s^0, a^0 & \text{otherwise} \end{cases} \quad (20)$$

where s^0 and a^0 are the zero-valued virtual observation and action vectors with the same dimensionality as the normal observation and action vectors; L represents the sequence length of the historical state information.

The AN and CN embedding GRU are updated by (21) and (22) [39], respectively:

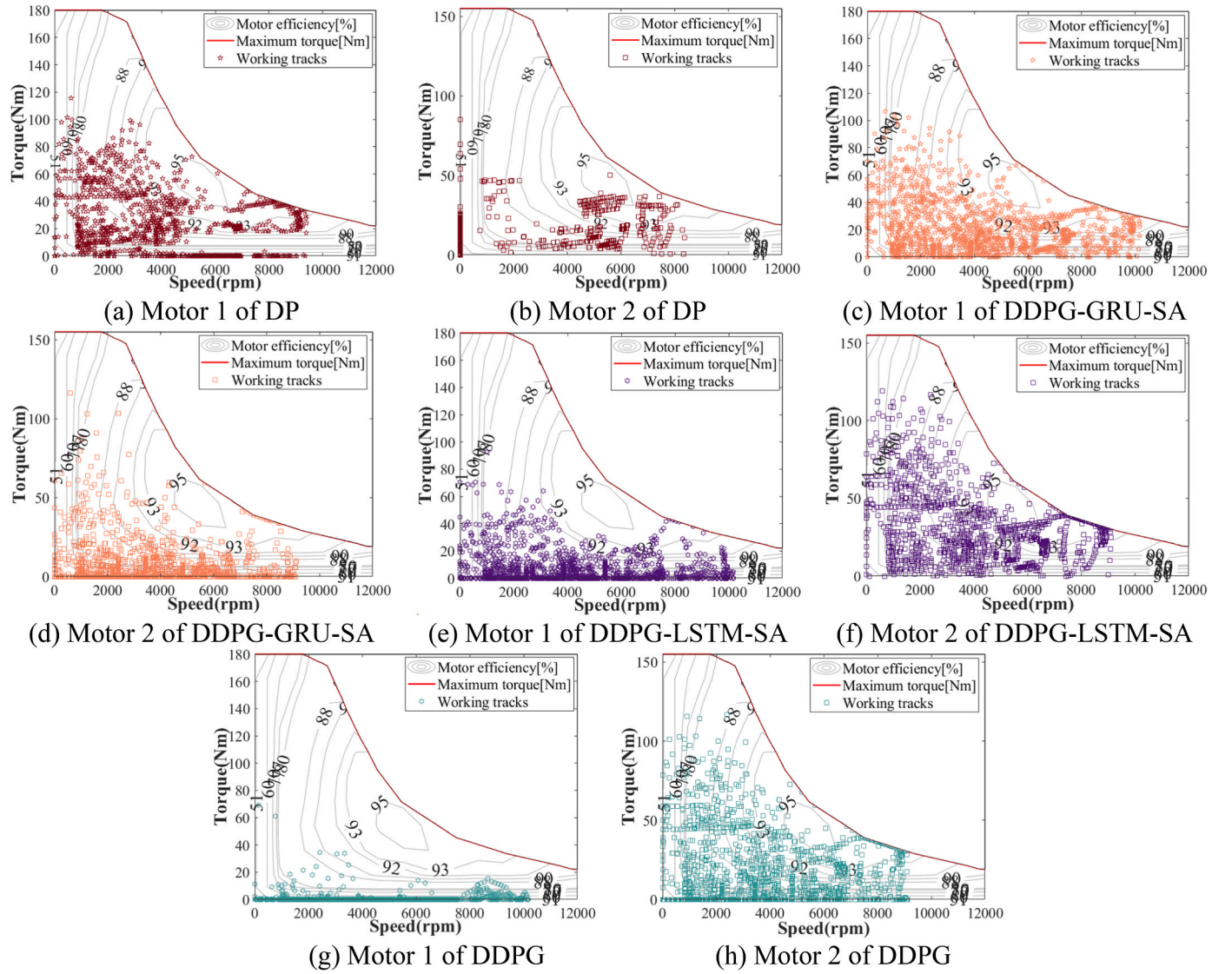


Fig. 6. The working tracks of two motors.

$$\begin{cases} Loss = E_{\{(h_t^L, a_t, s_t, r_t, s_{t+1})\}_{t=1}^M} (y_i - Q(s_t, a_t, h_t^L | \theta^Q))^2 \\ y_i = r_t + \gamma Q(s_{t+1}, \hat{a}, h_{t+1}^L | \theta^Q) \\ \hat{a} = \mu(s_{t+1}, h_{t+1}^L | \theta^\mu) + N \end{cases} \quad (21)$$

$$\nabla_{\theta^\mu} J = E_{\{(h_t^L, a_t, s_t, r_t, s_{t+1})\}_{t=1}^M} \begin{pmatrix} \nabla_{\theta^Q} Q(s_t, a_t, h_t^L | \theta^Q) \\ \nabla_{\theta^\mu} \mu(s_t, h_t^L | \theta^\mu) \end{pmatrix} \quad (22)$$

The GRU gives the DDPG agent the ability to process more state information, but each state information has a different impact on the reward, so it is necessary to distinguish the importance of these state information. Therefore, the SA mechanism is incorporated into the DDPG agent to prevent the GRU from losing crucial information during prolonged training. To summarize, the embedding of GRU and SA mechanism improves the learning efficiency of the original DDPG agent and boosts the reliability of its decision-making.

3.3. Configuring the reward, state & action spaces

Following its observation of the environment, the DRL agent acts, receives an instant reward, and transitions into a new state as a result of

its actions [35]. As a result, the three components mentioned above are essential to the design of DRL-based EMSs.

The chosen state space is intended to accurately depict the operational status of the BEVs. The vehicle's exterior motion is directly reflected in the velocity v and required torque T_d . The battery's state of charge is disclosed by the SOC. The state of health of the electronic components within the vehicle is indicated by the battery SOH, motor 1 SOH_{M1} , and motor 2 SOH_{M2} . State space is thus defined as:

$$State = [v, T_d, SOC, SOH, SOH_{M1}, SOH_{M2}] \quad (23)$$

The core idea of RL is that an agent learns the strategy with the highest expected reward in its interaction with the environment. Therefore, a well-designed reward function is crucial for RL-based EMS training. In our work, the EMS has three optimization objectives: 1) to reduce the electrical energy consumption of power battery; 2) to reduce the health degradation of the power battery; and 3) to reduce the health degradation of the dual motors. The action space and reward are determined in close relation to the objectives of the study, so they are set up as follows:

$$Action = [\sigma_{tor}] \quad (24)$$

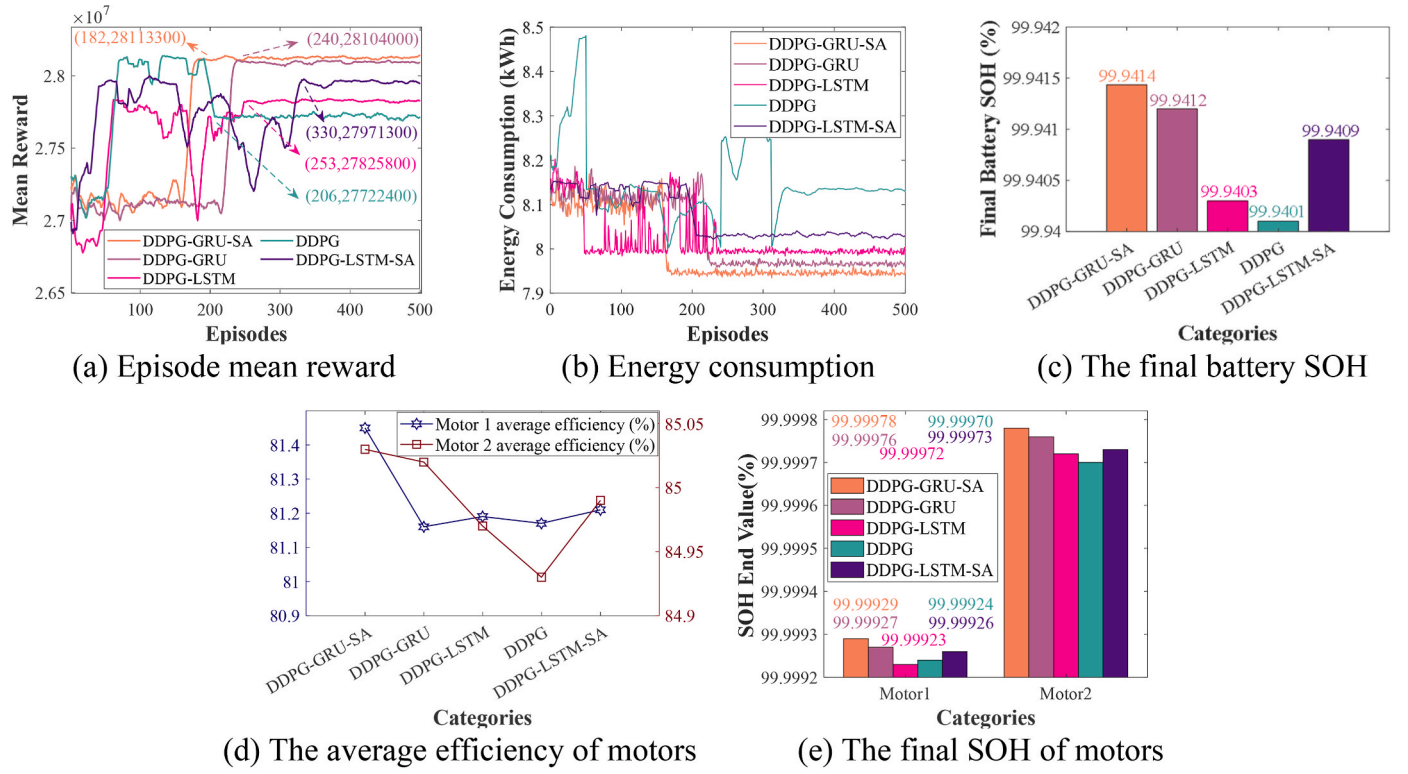


Fig. 7. Comparison of the performance of different EMSs.

$$Reward = \frac{1}{K_1 \Delta SOC + K_2 \Delta SOH + K_3 \Delta SOH_{M1} + K_4 \Delta SOH_{M2} + K_5} \quad (25)$$

where σ_{tor} is the torque distribution factor used to control the distribution of demand torque between motor 1 and motor 2; K_1, K_2, K_3, K_4 and K_5 are positive factors designed to control energy consumption and health decline, and their values are 1, 100, 1000, 1000 and 0.001, respectively.

Considering the security of the learning process, the following boundary conditions are given:

$$\left\{ \begin{array}{l} SOC_{min} \leq SOC \leq SOC_{max} \\ SOH_{batmin} \leq SOH \leq SOH_{batmax} \\ SOH_{M1 min} \leq SOH_{M1} \leq SOH_{M1 max} \\ SOH_{M2 min} \leq SOH_{M2} \leq SOH_{M2 max} \\ 0 \leq |N_{mot1}| \leq N_{M1 max} \\ 0 \leq |N_{mot2}| \leq N_{M2 max} \\ 0 \leq |T_{mot1}| \leq T_{M1 max} \\ 0 \leq |T_{mot2}| \leq T_{M2 max} \end{array} \right. \quad (26)$$

where SOC_{max} and SOC_{min} are set as 0.9 and 0.2, respectively; SOH_{batmax} , $SOH_{M1 max}$ and $SOH_{M2 max}$ are both set as 1; SOH_{batmin} , $SOH_{M1 min}$ and $SOH_{M2 min}$ are both set as 0.8; $N_{M1 max}$ and $N_{M2 max}$ are both set as 12000; $T_{M1 max}$ and $T_{M2 max}$ are set as 180 and 155, respectively.

The hyperparameters settings and the execution structure of the designed DRL-based EMS are presented in Table 2 and Fig. 3, respectively.

4. Simulation results and discussions

In this part, ablation experiments are conducted to confirm the effectiveness of the hypothesized GRU and SA methods. To enable

excellent generalization of the proposed EMS, a combined driving cycle covering low to medium to high speeds, as well as urban and suburban conditions is constructed, named mixed training cycle (it consists of WLTC and NEDC), as presented in Fig. 4(a). As seen in Fig. 4(b), the mixed driving cycle made up of UDDS, HWFET, CLTC-P, and FTP75 is chosen as the test cycle to confirm the adaptability of the suggested EMS. Moreover, as shown in Fig. 4(c), a traffic cycle constructed from real data is also used as the test cycle, which is derived from Ref. [40].

4.1. The study of GRU performance in EMSs

The software and hardware environments of the training EMSs are Windows 11, MATLAB R2023a, Intel i7-13700KF CPU, and RTX3060 GPU. Moreover, the state & action space, and reward function for all trained EMSs are identical.

Among the three EMSs, the one based on DDPG-GRU-SA converges at the fastest pace and has the largest mean reward, with a mean value of 28113300 at the 182nd episode, as shown in Fig. 5(a). However, the EMS based on DDPG-LSTM-SA converges at the 330th episode with the mean reward of 27971300, which is considerably slower than that of the EMS based on DDPG-GRU-SA. This phenomenon occurs since GRU speeds up convergence by simplifying the gate structure of the LSTM. The reward of DDPG-based EMS drops suddenly around 200th episode, gradually converging to a sub-optimal reward. This phenomenon may occur due to an overestimation of the Q-value, leading the agent to settle for a locally optimal solution.

Fig. 5(b) and (c) depict the battery SOC and SOH trajectories for the various EMSs, respectively. The two figures above demonstrate that compared to DDPG-LSTM-SA based and DDPG based EMSs, the suggested EMS based on DDPG-GRU-SA lowers the decline rate of both SOC and SOH. Fig. 5(d) shows that the proposed EMS has the highest final battery SOH except for the global optimal EMS, which proves that the proposed EMS guarantees a healthy battery performance. Fig. 5(e) and (f) respectively illustrate the SOH of motor 1 and motor 2, indicating that the proposed DDPG-GRU-SA based EMS closely matches DP-based

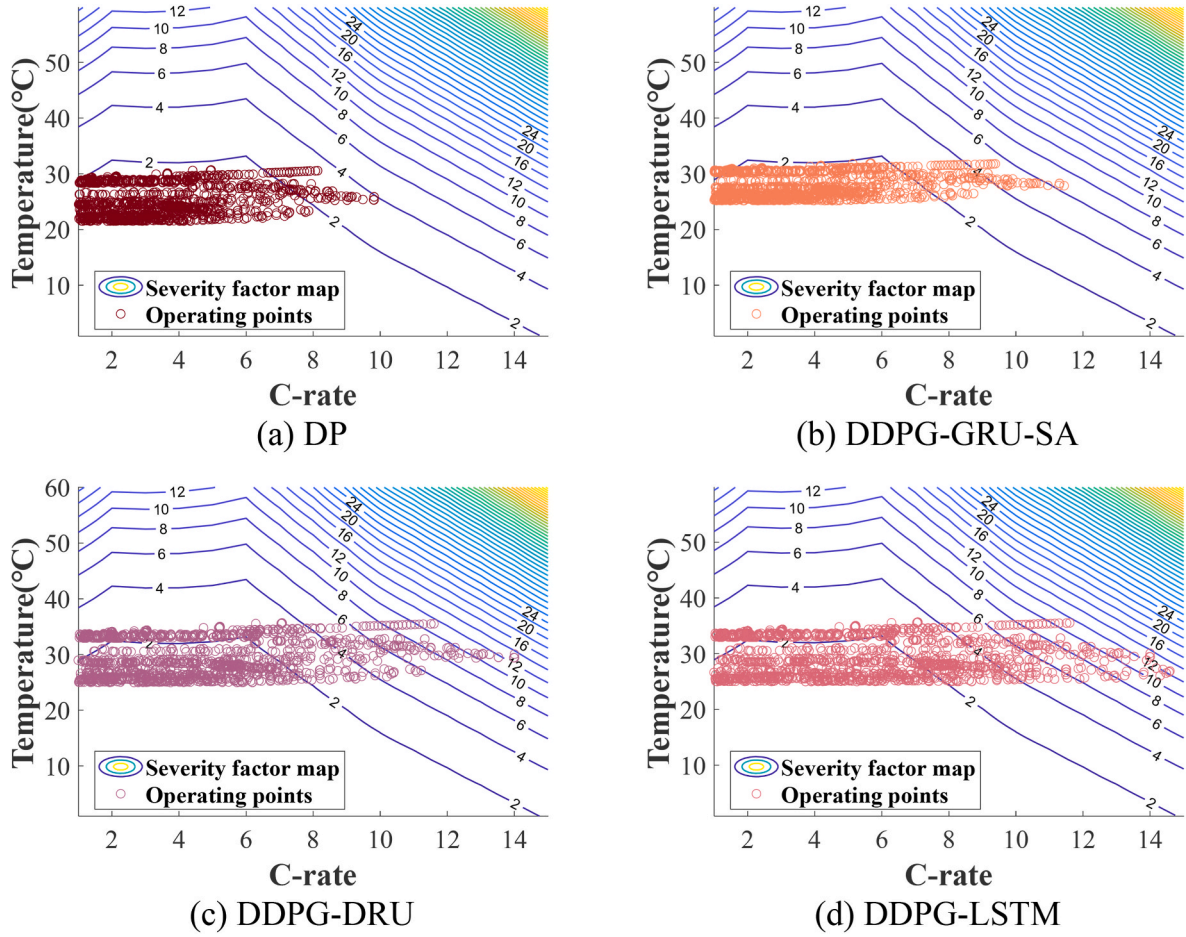


Fig. 8. Severity factor map of battery.

Table 4

Performance of EMSs under the mixed test cycle.

Initial SOC	EMSs	Electricity consumption (kWh)	Performance	Battery decline	Performance	Motors' decline	Performance
0.6	DP	13.33	100 %	0.98e-3	100 %	1.99e-4	100 %
	DDPG-GRU-SA	13.88	95.87 %	1.03e-3	94.90 %	2.07e-4	95.98 %
	DDPG-LSTM-SA	14.25	93.10 %	1.06e-3	91.84 %	2.12e-4	93.47 %
	DDPG-GRU	14.59	90.55 %	1.08e-3	89.80 %	2.16e-4	91.46 %
0.3	DP	13.71	100 %	0.99e-3	100 %	2.00e-4	100 %
	DDPG-GRU-SA	14.37	95.18 %	1.07e-3	91.92 %	2.13e-4	93.50 %
	DDPG-LSTM-SA	14.42	91.82 %	1.07e-3	91.92 %	2.14e-4	93.00 %
	DDPG-GRU	14.92	91.17 %	1.10e-3	88.89 %	2.21e-4	89.50 %

Table 5

Performance of EMSs under the actual test cycle.

Initial SOC	EMSs	Electricity consumption (kWh)	Performance	Battery decline	Performance	Motors' decline	Performance
0.6	DP	6.75	100 %	4.95e-4	100 %	1.01e-5	100 %
	DDPG-GRU-SA	7.11	94.67 %	5.25e-4	93.94 %	1.07e-5	94.06 %
	DDPG-LSTM-SA	7.29	92.00 %	5.31e-4	92.73 %	1.09e-5	92.08 %
	DDPG-GRU	7.42	90.07 %	5.40e-4	90.90 %	1.13e-5	88.12 %
0.3	DP	6.89	100 %	5.07e-4	100 %	1.03e-5	100 %
	DDPG-GRU-SA	7.30	94.05 %	5.38e-4	93.89 %	1.10e-5	93.20 %
	DDPG-LSTM-SA	7.45	91.87 %	5.52e-4	91.12 %	1.12e-5	91.26 %
	DDPG-GRU	7.58	89.98 %	5.59e-4	89.74 %	1.15e-5	88.35 %

EMS in maintaining motors' health. Table 3 provides a quantitative analysis of the above phenomenon. Notably, the value of motors' health decline in the Table 3 is the sum of two motors.

Using the globally optimal DP-based EMS as a baseline, as the values

summarized in Table 3. In respect of electricity consumption, DDPG-GRU-SA-based EMS, DDPG-LSTM-SA-based EMS, and DDPG-based EMS can reach 96.88 %, 95.85 %, and 94.55 % of DP-based EMS, respectively. Furthermore, the battery and motor deteriorations are

substantially suppressed by proposed EMS, reaching 96.83 % and 97.83 % of the baseline, respectively. The DDPG-based EMS shows the lowest performance in maintaining battery and motor health, achieving only 94.54 % and 92.40 % of the DP-based EMS, respectively, consistent with its training performance. The running time of the EMS developed on DDPG-GRU-SA is 31520 s under 500 episodes, which is much lower than the 35230 s for DDPG-LSTM-SA-based EMS and the 40123 s for EMS based on original DDPG. It's important to note that the baseline operates for the longest running time because of the pursuit of optimality.

Fig. 6 shows the two motors' working tracks under various EMSs. As seen in Fig. 6(a) and (b), respectively, the DP-based EMS allows the operating points of motors 1 and 2 to be more concentrated in the high-efficiency zone. Fig. 6(c) and (d) depict the working tracks of the two motors for the proposed EMS based on DDPG-GRU-SA, respectively, and they exhibit a high degree of consistency with the DP-based EMS. Fig. 6(e) shows that the EMS based on DDPG-LSTM-SA makes motor 1 hardly operate in the region of efficiency greater than 92 %. Comparing with Fig. 6(b) and (d), motor 2 in Fig. 6(f) has more dense working points (Fig. 6(h) shows a similar working situation), which means that motor 2 runs more times in the low-efficiency region. Fig. 6(g) illustrates that the DDPG-based EMS causes motor 1 to operate in the lower-than-80 % efficiency range. One possible reason is that the DDPG agent is unable to observe both historical and current environmental information (An agent utilizing DDPG and incorporating either GRU or LSTM can achieve this task). The missing environmental information hinders the DDPG agent from exploring the optimal policy.

4.2. The study of self-attention performance in EMSs

Several ablation tests are carried out in this section to confirm the SA mechanism's functionality. As illustrated in Fig. 7(a), the DDPG-GRU-based and DDPG-LSTM-based EMSs without the integrating SA mechanism converge in the 240th and 253rd episodes, respectively. The convergence speed and reward of the above two EMSs are lower than that of proposed EMS based on DDPG-GRU-SA. This is because SA mechanism ensures that the GRU does not discard critical information during long training processes. Therefore, the learning efficiency of the DDPG agent is improved. It can be noted from Fig. 7(b) that electrical energy consumption of DDPG-GRU-SA-based EMS keeps decreasing and quickly stabilizes to the range of about 7.95 kWh. It is important to note that energy consumption of the DDPG-based EMS fluctuates dramatically at the early part of the exploration, and it also consumes significantly more electrical energy than the other four EMSs when it stabilizes.

Fig. 7(c) shows the proposed DDPG-GRU-SA-based EMS has the best battery health performance with a final SOH of 99.9414 %. Significantly, DDPG-based EMS has the worst battery health performance with a final SOH of 99.9401 %, which is similar with the performance of DDPG-LSTM. Fig. 7(d) analyzes the average efficiency of motors for different EMSs. The proposed EMS makes the two motors have the highest average motor efficiency of about 81.48 % and 85.03 %, respectively. The minimum average efficiency of motor 2 in DDPG-based EMS is around 84.92 %, while motor 1 in DDPG-GRU-based EMS has a minimum average efficiency of roughly 81.15 %. As described in Section 2, motor degradation is closely related to motor efficiency. Thus, the proposed DDPG-GRU-SA based EMS has the highest terminal SOH of motors, which are 99.99929 % and 99.99978 % respectively, and this phenomenon is depicted in Fig. 7(e).

Reference [31] suggests that when the battery operating temperature exceeds 30 °C, the life of the lithium-ion battery is shortened by approximately two months for each degree of elevation. Additionally, Power cell degradation is affected by high C-rate besides high temperatures [32,41]. Fig. 8 shows the operating trajectories of the power battery at a certain temperature and C-rate. DP-based EMS as a globally optimal strategy enables the battery to operate in a healthy environment where the temperature rarely exceeds 30 °C and the C-rate rarely

exceeds 10, as shown in Fig. 8(a). Fig. 8(b) presents the proposed DDPG-GRU-SA-based EMS is comparable to DP-based EMS in terms of its effectiveness in controlling the temperature, but the C-rate is slightly increased to the range of about 12. By observing Fig. 8(c) and (d), it can be noticed that the EMSs without incorporating the SA mechanism (i.e., DDPG-GRU-based and DDPR-LSTM-based EMSs) not only raise the operating temperature to the range of about 35 °C but also raises the C-rate to the range of about 14. One possible reason for this phenomenon is that the SA mechanism achieves information filtering and fusion by distinguishing the importance of GRU output information. On this basis, the SA mechanism increases the reliability of DDPG agents' decisions.

4.3. Adaptability analysis of the proposed EMSs

In the real world, the actual driving conditions of vehicles are unlikely to be the same as the training cycle. Thus, following offline training, an online test is necessary to validate the applicability of the suggested EMS. To accomplish the above objective, DRL agents after convergence are extracted and the test cycles are fed directly into the converged agents.

The preliminary SOC is set to 0.60 and 0.30, respectively, to simulate the full and short battery power in starting phase, which is distinguished from the excess battery power during training. From the Table 4, when the initial SOC is 0.6, in comparison with the DDPG-LSTM-SA based EMS, EMS based on DDPG-GRU-SA achieved improvements of 2.77 %, 3.06 %, and 2.51 % in the electricity consumption, battery decline, and motors' decline respectively. The suggested EMS outperformed the DDPG-GRU-based EMS in the aforementioned three optimization objectives by 5.32 %, 5.10 %, and 4.52 %, in that order. When the initial SOC is 0.3, the electricity consumption, battery decline, and motors' decline of the proposed EMS reach 95.18 %, 91.92 %, and 93.50 % of DP-based EMS, respectively. From the Table 5, when the initial SOC is 0.6, DDPG-GRU-SA based EMS narrows the gap with DP-based EMS to 5.33 %, 6.06 %, and 5.94 % in the electricity consumption, battery decline, and motors' decline respectively. When the initial SOC is 0.3, the suggested EMS outperformed the DDPG-LSTM-SA-based EMS in the three optimization objectives by 2.18 %, 2.77 %, and 1.94 %, in that order. The results of comparisons demonstrate that the proposed EMS provides better adaptability compared to the EMSs that does not employ all proposed optimization techniques.

5. Conclusion

This study designs the EMS based on DDPG-GRU-SA for the BEV. The main work can be summarized as follows: 1) To improve the economy requirement of the BEV, electrical energy consumption, and the health of the battery and motors are simultaneously introduced into the optimization objectives. 2) Aiming at the inability of original DDPG to capture historical state information, GRU is embedded into its actor-critic networks, which can solve computational inefficiency of LSTM. 3) In addition, the ability of the DDPG agent to perceive and learn state information is enhanced by introducing the SA mechanism. After intensive simulation, the following meaningful conclusions can be drawn: 1) The proposed EMS speeds up the agent learning process and shortens the converged episodes to 182nd. 2) The proposed EMS achieves 96.88 %, 96.83 %, and 97.83 % of DP-based EMS in terms of electricity consumption, battery degradation, and motors' degradation, respectively. 3) Under the test cycles, the proposed EMS outperforms the compared EMSs regardless of the initial SOC, proving its superior adaptability.

The DDPG-GRU-SA based EMS has limitations, including manual sequence length L selection impacting computational efficiency and the need for hardware-in-the-loop testing to validate real-time capabilities.

CRediT authorship contribution statement

Changcheng Wu: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Jiankun Peng:** Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Hongwen He:** Writing – review & editing, Writing – original draft. **JIAGENG RUAN:** Visualization, Validation, Conceptualization. **Jun Chen:** Methodology, Investigation, Conceptualization. **Chunye Ma:** Methodology, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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